

US Patent & Trademark Office

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United States Patent Application Publication

20250273443

Kind Code

A1

Publication Date

August 28, 2025

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SUBSTRATE PROCESSING APPARATUS INCLUDING EDGE BIAS SUPPLY PORTION

Abstract

A substrate processing apparatus includes a chamber, a support member disposed inside the chamber and including a bias electrode and an edge bias electrode, a bias supply portion electrically connected to the bias electrode, and an edge bias supply portion electrically connected to the edge bias electrode.

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Family ID: 1000008125244

Appl. No.: 18/807142

Filed: August 16, 2024

Foreign Application Priority Data

KR

10-2024-0028788

Feb. 28, 2024

Publication Classification

Int. Cl.: H01J37/32 (20060101); H01L21/683 (20060101)

U.S. Cl.:

CPC H01J37/32697 (20130101); H01J37/32183 (20130101); H01J37/32642 (20130101); H01J37/32715 (20130101); H01J37/32935 (20130101); H01L21/6833 (20130101); H01J2237/334 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to and the benefit of Korean Patent Application No. 10-2024-0028788 filed in the Korean Intellectual Property Office on Feb. 28, 2024, the entire contents of which are herein incorporated by reference.

BACKGROUND

(a) Technical Field

[0002] The present disclosure relates to a substrate processing apparatus, and more particularly to a substrate processing apparatus including an edge bias supply portion.

(b) Discussion of Related Art

[0003] The manufacture of semiconductor devices typically involves various processes performed on a substrate. These processes may include photolithography, etching, ashing, ion implantation, thin film deposition, and cleaning. These processes may be performed on the substrate to form a desired pattern. For example, the etching process may remove a portion of a film formed on the substrate, where a remaining portion of the film forms the pattern. The etching process may be a wet etching or a dry etching.

[0004] Dry etching may be performed to remove a portion of the film by bombarding the film with plasma. Typically, to form plasma, an electromagnetic field may be formed in an inner space of a chamber, and the electromagnetic field may excite a process gas provided in the chamber into a plasma state.

[0005] Plasma may refer to an ionized gas state composed of ions, electrons, and radicals. Plasma may be generated by high temperatures, strong electric fields, or strong radio frequency (RF) electromagnetic fields.

SUMMARY OF THE INVENTION

[0006] The present disclosure describes a substrate processing apparatus that may process a substrate in which a state of a sheath (of a plasma) may be effectively adjusted.

[0007] However, objects of the present disclosure are not limited thereto, and may be extended in various ways within the spirit and scope of the present disclosure.

[0008] An embodiment provides a substrate processing apparatus including: a chamber; a support member disposed inside the chamber and including a bias electrode and an edge bias electrode; a bias supply portion electrically connected to the bias electrode; and an edge bias supply portion electrically connected to the edge bias electrode, wherein the edge bias supply portion applies a coupling control signal having a frequency corresponding to a coupling signal, and the coupling signal is generated at the edge bias electrode by coupling with a bias signal generated at the bias electrode.

[0009] Another embodiment provides a substrate processing apparatus including: a chamber; a support member disposed inside the chamber and including a bias electrode and an edge bias electrode; a bias supply portion electrically connected to the bias electrode; an edge bias supply portion electrically connected to the edge bias electrode; and a measurement member electrically

connected to the edge bias electrode, wherein the measurement member measures a coupling signal generated at the edge bias electrode by coupling with a bias signal generated at the bias electrode.

[0010] Another embodiment provides a substrate processing apparatus including: a chamber; a support member disposed inside the chamber and including a bias electrode and an edge bias electrode; a bias supply portion electrically connected to the bias electrode; an edge bias supply portion electrically connected to the edge bias electrode; and a measurement member electrically connected to the edge bias electrode, wherein the measurement member measures a coupling signal generated at the edge bias electrode by coupling with a bias signal generated at the bias electrode, and the edge bias supply portion applies a coupling control signal having a frequency corresponding to a reflected signal to generate a synthetic signal in which at least one of an amplitude or a phase of the synthetic signal is different than an amplitude or a phase of the reflected signal, wherein the reflected signal is a portion of the coupling signal traveling in a direction from the edge bias supply portion toward the edge bias electrode.

[0011] According to some embodiments, it is possible to provide a substrate processing device that may process a substrate in a state in which a state of a sheath is effectively controlled.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1 illustrates a substrate processing apparatus according to an embodiment.

[0013] FIG. 2 illustrates an upper portion of a support member of FIG. 1.

[0014] FIG. 3 illustrates a coupling signal generated at an edge bias electrode.

[0015] FIG. 4 illustrates a relationship between a coupling control signal and a reflected signal applied by an edge bias supply portion according to an embodiment.

[0016] FIG. 5 illustrates a relationship between a coupling control signal and a reflected signal applied by an edge bias supply portion according to another embodiment.

[0017] FIG. 6 and FIG. 7 respectively illustrate a relationship between a coupling control signal and a reflected signal applied by an edge bias supply portion according to another embodiment.

[0018] FIG. 8 illustrates a relationship between an incident signal and a synthetic signal according to an embodiment.

[0019] FIG. 9 illustrates a relationship between an incident signal and a synthetic signal according to another embodiment.

[0020] FIG. 10 and FIG. 11 respectively illustrate a relationship between an incident signal and a synthetic signal according to another embodiment.

[0021] FIG. 12 illustrates an edge bias supply portion according to an embodiment.

[0022] FIG. 13 illustrates an upper portion of a support member according to another embodiment.

[0023] FIG. 14 illustrates an upper portion of a support member according to another embodiment.

[0024] FIG. 15 illustrates an upper portion of a support member according to another embodiment.

[0025] FIG. 16 illustrates a substrate processing apparatus according to another embodiment.

[0026] FIG. 17 illustrates a substrate processing apparatus according to another embodiment.

DETAILED DESCRIPTION

[0027] Embodiments of the present disclosure will be described more fully hereinafter with reference to the accompanying drawings. The present disclosure may be implemented in various different forms and is not limited to embodiments provided herein. Embodiments may be modified in various different ways, all without departing from the spirit or scope of the present disclosure. In order to clearly describe aspects of the present disclosure, parts or portions that may be irrelevant to the description may be omitted.

[0028] In the drawings, the size and thickness of each element may be arbitrarily illustrated for ease of description. In the drawings, the thicknesses of layers, films, panels, regions, or areas may

be exaggerated for clarity. In the drawings, for ease of description, the thicknesses of some layers and areas may be exaggerated.

[0029] It will be understood that when an element such as a layer, film, region, area, or substrate is referred to as being “on” or “above” another element, it may be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” another element, there may be no intervening elements present. Further, in the specification, the word “on” or “above” means disposed on or below the object portion, and does not necessarily mean disposed on the upper side of the object portion based on a gravitational direction.

[0030] In addition, unless explicitly described to the contrary, the word “comprise” and variations such as “comprises” or “comprising” may be understood to imply the inclusion of stated elements but not the exclusion of any other elements.

[0031] Further, throughout the specification, the phrase “in a plan view” or “on a plane” means viewing a target portion from the top, and the phrase “in a cross-sectional view” or “on a cross-section” means viewing a cross-section formed by vertically cutting a target portion from the side.

[0032] FIG. 1 illustrates a substrate processing apparatus 1 according to an embodiment.

[0033] Referring to FIG. 1, the substrate processing apparatus 1 according to an embodiment may include a chamber 10, a support member 20, and a plasma excitation member 50.

[0034] The substrate processing apparatus 1 may process a substrate using plasma. For example, the substrate processing apparatus 1 may perform an etching process using excited plasma. The substrate may be a wafer for manufacturing a semiconductor device. The chamber 10 may have a cylindrical shape. For example, the chamber 10 may include one or more sidewalls connected between a top and a bottom. However, a shape of the chamber 10 is not limited thereto. The chamber 10 may define a process space PS within which a substrate processing process may be performed. The chamber 10 may include an internal process space PS. The internal process space may be a sealed space. An opening 11 may be disposed at a sidewall of the chamber 10. The opening 11 may be opened and closed. The opening 11 may be sealed by a door 12. The chamber 10 may be made of a metallic material. For example, the chamber 10 may be made of an aluminum material. The chamber 10 may be electrically grounded.

[0035] The support member 20 may be disposed inside the chamber 10. The support member 20 may be disposed at a lower portion of the process space PS. The support member 20 may support the substrate. The support member 20 may fix the substrate using an electrostatic force.

[0036] The support member 20 may have a cylindrical shape. However, a shape of the support member 20 is not limited thereto. The support member 20 may define an inner area. The support member 20 may include a fixing electrode 21, a heating member 22, a lower electrode 23, and an edge bias electrode 27.

[0037] The fixing electrode 21 may allow an electrostatic force for fixing the substrate to be generated. The fixing electrode 21 may be disposed in the inner area of the support member 20. A fixing supply portion 31 may be connected to the support member 20. The fixing supply portion 31 may be electrically connected to the fixing electrode 21. The fixing supply portion 31 may include a direct current (DC) power source. An electrostatic force may be generated between the fixing electrode 21 and the substrate by a voltage applied by the fixing supply portion 31, and the electrostatic force may cause the substrate to be fixed to the support member 20.

[0038] The heating member 22 may generate heat. The heating member 22 may generate heat for heating the substrate. The heating member 22 may be disposed in an inner area of the support member 20. The heating member 22 may generate heat through resistance heating when power is applied thereto, which may cause the substrate disposed on the support member 20 to be heated. The heating member 22 may be made of a conductive material. The heating member 22 may be disposed below the fixing electrode 21.

[0039] A heating supply portion 32 may be connected to the support member 20. The heating

supply portion **32** may be electrically connected to the heating member **22**. The heating member **22** may be resistance-heated by power supplied by the heating supply portion **32**, and the substrate disposed on the support member **20** may be heated.

[0040] The lower electrode **23** may allow energy for plasma excitation to be applied to the process space PS. The lower electrode **23** may be disposed in the inner area of the support member **20**. The lower electrode **23** may be made of a conductive material. The lower electrode **23** may be made of a metallic material.

[0041] An excitation supply portion **33** may be connected to the support member **20**. The excitation supply portion **33** may be electrically connected to the lower electrode **23**. The excitation supply portion **33** may be provided as a high-frequency power source that may generate high-frequency power. The excitation supply portion **33** may be provided as an RF power source. The excitation supply portion **33** may provide power for plasma excitation. The excitation supply portion **33** may be connected to the lower electrode **23** through a matcher **34**. The matcher **34** may be disposed between the excitation supply portion **33** and the lower electrode **23** to perform an RF impedance matching operation.

[0042] In addition, the excitation supply portion **33** electrically connected to the lower electrode **23** may be omitted, and the lower electrode **23** may be electrically grounded.

[0043] A bias supply portion **35** may be connected to the support member **20**. The bias supply portion **35** may be electrically connected to the lower electrode **23**. Accordingly, the lower electrode **23** may function as a bias electrode **23**. The bias supply portion **35** provides power for bias. The bias may generate an electron sheath (hereinafter “sheath”) that may be a thin region of negative charge that acts to accelerate electrons out of the plasma and reflect ions back into the plasma.

[0044] A state of the sheath may be adjusted in an area adjacent to an upper surface of the support member **20** by the power supplied by the bias supply portion **35**. As the sheath is adjusted, a concentration state of plasma on the substrate and an incident state of ions on the substrate may be adjusted. Other properties of the plasma may also be adjusted by the bias supply portion **35**. A frequency of the bias supply portion **35** may be provided to be lower than that of the excitation supply portion **33**. For example, the frequency of the excitation supply portion **33** may be about 50 megahertz (MHz) to hundreds of MHz, and the frequency of the bias supply portion **35** may be about 0.1 MHz to about 15 MHz.

[0045] The edge bias electrode **27** may allow a sheath state of an area adjacent to an upper surface of the edge area of the support member **20** to be adjusted. The edge bias electrode **27** may be disposed at an upper end portion of the edge area of the support member **20**.

[0046] An edge bias supply portion **40** may be connected to the support member **20**. The edge bias supply portion **40** may be electrically connected to the edge bias electrode **27**. A state of the sheath may be adjusted in an area adjacent to the edge area of the upper surface of the support member **20** by the power supplied by the edge bias supply portion **40**. As the sheath is adjusted, a concentration state of plasma on an edge area of the substrate and an incident state of ions on an edge area of the substrate may be adjusted. Other properties of the plasma may also be adjusted by the edge bias supply portion **40**.

[0047] The edge bias supply portion **40** may be connected to the support member **20** through an impedance control member **42**. The edge bias supply portion **40** may be electrically connected to the edge bias electrode **27** through the impedance control member **42**. The impedance control member **42** may be disposed between the edge bias supply portion **40** and the edge bias electrode **27**. The impedance control member **42** may control a coupling signal generated at the edge bias electrode **27**. For example, in the support member **20**, a signal (hereinafter, a bias signal) generated by the bias electrode **23** may exist in an area adjacent to the edge bias electrode **27**. When the edge bias electrode **27** is coupled with a bias signal in an area adjacent to the edge bias electrode **27**, a coupling signal may be generated in the edge bias electrode **27**. The impedance control member **42**

may be electrically connected to the edge bias electrode **27** and may adjust the coupling signal generated in the edge bias electrode **27**.

[0048] A measurement member **41** may be connected to the support member **20**. The measurement member **41** may be electrically connected to the edge bias electrode **27**. The edge bias supply portion **40** may be connected to the edge bias electrode **27** through the measurement member **41**. Accordingly, the measurement member **41** may be disposed between the edge bias supply portion **40** and the edge bias electrode **27**. The measurement member **41** may be disposed between the impedance control member **42** and the edge bias electrode **27**. Further, the measurement member **41** may be disposed closer to the edge bias electrode **27** than the impedance control member **42**. The measurement member **41** may measure a coupling signal generated at the edge bias electrode **27**.

[0049] The plasma excitation member **50** may allow energy for excitation of plasma to be applied to the process space PS. The lower surface of the plasma excitation member **50** may be disposed to face an inner portion of the chamber **10** and a volume of the process space PS. The plasma excitation member **50** may be disposed inside the chamber **10**. For example, the plasma excitation member **50** may be manufactured separately from the chamber **10** and connected to the chamber **10**. Alternatively, the plasma excitation member **50** may be integrally provided with an upper structure of the chamber **10**. For example, the plasma excitation member **50** may be disposed on a lower surface of the top of the chamber **10**. In other words, the upper structure of the chamber **10** may function as the plasma excitation member **50**.

[0050] The plasma excitation member **50** may be disposed in the upper portion of the process space PS. The plasma excitation member **50** may be made of a conductive material. An area of the plasma excitation member **50** may be less than an area of the top of the chamber **10**. The plasma excitation member **50** may be disposed to face the support member **20** in a vertical direction.

[0051] A process gas introduced into the chamber **10** may be excited into a plasma state by an electromagnetic field generated inside the chamber **10**. Specifically, the process gas may be excited into plasma by a capacitively coupled plasma (CCP) source. The capacitively coupled plasma source may include an upper electrode and a lower electrode **23**. The upper electrode and the lower electrode **23** may be disposed in the vertical direction to face each other. By applying high frequency power to at least one of the upper electrode and the lower electrode **23**, an electromagnetic field may be formed in the space between the upper electrode and the lower electrode **23**, and the process gas supplied to the space may be excited in a plasma state. The upper electrode may be the plasma excitation member **50**.

[0052] The excitation supply portion **33** may be connected to only one of the upper electrode and the lower electrode **23**. For example, the upper electrode may be grounded, and the excitation supply portion **33** may be connected the lower electrode **23**. Alternatively, the lower electrode **23** may be grounded, and the excitation supply portion **33** may be connected the upper electrode. In another example, the excitation supply portion **33** may be connected to the upper electrode and the lower electrode **23**. FIG. 1 illustrates an example case in which the excitation supply portion **33** is connected to the lower electrode **23**.

[0053] FIG. 2 illustrates an upper portion of the support member **20** of FIG. 1.

[0054] Referring to FIG. 2, the support member **20** may include a support plate **200**, a focus ring **210**, and a support ring **220**.

[0055] The support plate **200** may be disposed in an upper area of the support member **20**.

Accordingly, at least a portion of the upper surface of the support member **20** may be formed by an upper surface of the support plate **200**. The support plate **200** may have a plate structure with a predetermined thickness. An outer surface of the support plate **200** may be circular. For example, the support plate **200** may have a circumference. The outer surface of the support plate **200** may be made of a dielectric substance. A substrate may be disposed on the upper surface of the support plate **200**. An area of the upper surface of the support plate **200** may greater than or equal to an area

of the substrate.

[0056] The fixing electrode **21** may be disposed inside the support plate **200**. The heating member **22** may be disposed inside the support plate **200**. The lower electrode **23** may be disposed below the support plate **200**.

[0057] The focus ring **210** may be disposed to be an upper edge portion of the support member **20**. The focus ring **210** may define an upper outer circumference of the support member **20**.

[0058] At least a portion of the focus ring **210** may be disposed on the support plate **200**. The focus ring **210** may be disposed on an outer periphery of an upper surface of the support plate **200**. The outer periphery of the upper surface of the support plate **200** may be recessed. For example, the outer periphery of the upper surface of the support plate **200** may have a lower height than an inner area of the upper surface of the support plate **200**.

[0059] A portion of an upper surface of the focus ring **210** may be recessed. For example, a portion of an upper surface of the focus ring **210** adjacent to the support plate **200** may include a step. The step of the focus ring **210** may overlap the outer periphery of the upper surface of the support plate **200**.

[0060] The focus ring **210** may be provided in a ring shape. The focus ring **210** may define a circumference of the support member **20**.

[0061] The upper surface of the focus ring **210** may be stepped, and an outer portion of the upper surface of the focus ring **210** may be higher than an inner portion of the upper surface of the focus ring **210**. The upper inner portion of the focus ring **210** may be disposed at a height corresponding to the upper surface of the support plate **200**. The inner portion of the upper surface of the focus ring **210** may have a height coplanar with the inner area of the upper surface of the support plate **200**. When the substrate is disposed on the support member **20**, the inner portion of the upper surface of the focus ring **210** and the inner area of the upper surface of the support plate **200** may be disposed below the substrate. The outer portion of upper surface of the focus ring **210** may be disposed to surround the substrate.

[0062] The focus ring **210** may improve a uniformity of a density distribution of plasma. The focus ring **210** may be worn during use. As the wear of the focus ring **210** increases, the performance of controlling the density distribution of plasma may deteriorate. Accordingly, the focus ring **210** may be replaced after being used for a certain period of time, a certain number of times, or after a measured wear. However, aspects of the present disclosure are not limited thereto, and the focus ring **210** may be freely replaced.

[0063] The support ring **220** may define an outer circumference of the support member **20**. The outer circumference of the support member **20** may be disposed below the upper outer circumference of the support plate **200** defined by the focus ring **210**. The support ring **220** may surround at least a portion of the outer circumference of the support plate **200**. The support ring **220** may have a ring structure in which at least a portion thereof has a height in the vertical direction. Accordingly, an area of the support plate **200** surrounded by the support ring **220** may be isolated from the internal space of the chamber **10**. The support ring **220** may be made of an insulating material. The outer circumference of the support plate **200** may be provided to be surrounded by the support ring **220** and the focus ring **210**, and the outer circumference of the support plate **200** may be isolated from the internal space of the chamber **10**.

[0064] An upper surface of the support ring **220** may be disposed at a height corresponding to the outer edge area of the upper surface of the support plate **200**. The support ring **220** may be disposed below the focus ring **210**. At least a portion of an area of the focus ring **210** may be disposed on the upper surface of the support ring **220**. At least a portion of an area of the lower surface of the focus ring **210** may face the upper surface of the support ring **220** in a vertical direction.

[0065] The edge bias electrode **27** may be disposed below the focus ring **210**. The edge bias electrode **27** may be disposed to face the focus ring **210** in the vertical direction. The edge bias electrode **27** may be disposed inside the support ring **220**. The edge bias electrode **27** may be

disposed at an upper end portion of the support ring **220**. The edge bias electrode **27** may be disposed at a height between the upper surface and the lower surface of the support plate **200** in the vertical direction.

[0066] The focus ring **210** may be coupled to the edge bias electrode **27**. Accordingly, a state of the sheath in an area adjacent to the upper surface of the focus ring **210** may be adjusted by a signal generated at the edge bias electrode **27**.

[0067] FIG. **3** illustrates a coupling signal generated at the edge bias electrode **27**.

[0068] The horizontal axis may represent time, and the vertical axis may represent voltage, which may represent the magnitude of the signal.

[0069] Referring to FIG. **3**, the coupling signal may include an incident signal and a reflected signal.

[0070] A bias signal generated from the bias electrode **23** may be applied to the support plate **200**. The bias signal may cause the state of the sheath on the upper surface of the support plate **200** to be adjusted. Depending on the state of the sheath, the concentration state of plasma on the substrate and the incident state of ions on the substrate may be adjusted. Other properties of the plasma may also be adjusted by the bias signal. The edge bias electrode **27** may be coupled to the bias signal. The edge bias electrode **27** may be directly coupled to the bias signal of the support plate **200** or may be coupled to the bias signal of the support plate **200** through the focus ring **210**. By coupling the bias signal to the edge bias electrode **27**, the coupling signal may be generated at the edge bias electrode **27**.

[0071] The incident signal may be directly generated at the edge bias electrode **27** by coupling with the bias signal. The incident signal may travel in a direction from the edge bias electrode **27** to the edge bias supply portion **40**.

[0072] A portion of the incident signal may be reflected. For example, a reflected signal may be generated when the incident signal is reflected from the impedance control member **42**. The reflected signal may travel in a direction from the edge bias supply portion **40** to the edge bias electrode **27**. The incident signal and the reflected signal are illustrated in FIG. **3**.

[0073] Accordingly, the coupling signal, which is a synthetic signal of the incident signal and the reflected signal, may be generated at the edge bias electrode **27**. As the coupling signal is generated according to the coupling of the bias signal and the edge bias electrode **27**, the coupling signal may be generated even when the edge bias supply portion **40** is in an off state. The coupling signal may affect the state of the sheath on the upper surface of the focus ring **210**.

[0074] The measurement member **41** may be electrically connected to the edge bias electrode **27**, and may measure the coupling signal. The measurement member **41** may measure the incident signal and the reflected signal separately. The measurement member **41** may be disposed between the edge bias electrode **27** and the edge bias supply portion **40** to measure the reflected signal traveling in a direction from the edge bias supply portion **40** toward the edge bias electrode **27**. The measurement member **41** may be disposed between the edge bias electrode **27** and the impedance control member **42** to measure the reflected signal generated when the incident signal is reflected from the impedance control member **42**.

[0075] The edge bias supply portion **40** may apply the coupling control signal to the edge bias electrode **27**. The coupling control signal applied to the edge bias electrode **27** may control the coupling signal generated at the edge bias electrode **27**. The coupling control signal applied by the edge bias supply portion **40** may have a frequency corresponding to the coupling signal. The coupling control signal applied by the edge bias supply portion **40** may have a frequency corresponding to the reflected signal. The edge bias supply portion **40** may apply a coupling control signal having a preset phase difference from the reflected signal. That is, by using the reflected signal measured by the measurement member **41**, the edge bias supply portion **40** may apply a coupling control signal having a preset phase difference from the reflected signal.

[0076] FIG. **4** illustrates a relationship between the coupling control signal applied by the edge bias

supply portion **40** and the reflected signal according to an embodiment.

[0077] The reflected signal is shown in graph **401**, and the coupling control signal is shown in graph **402**. The horizontal axis may represent time, and the vertical axis may represent voltage, which may be the magnitude of the signal.

[0078] Referring to FIG. **4**, the edge bias supply portion **40** may apply a coupling control signal having a phase that destructively interferes with the reflected signal. Destructive interference may occur when different waves interact so that the waves combine to produce a resultant wave with a smaller displacement. That is, the edge bias supply portion **40** may apply a coupling control signal having a phase difference (A) of about 180° relative to the reflected signal. Accordingly, the coupling control signal may destructively interfere with the reflected signal and, a synthetic signal may be generated in which the coupling control signal and the reflected signal are synthesized.

[0079] An amplitude of the coupling control signal may be larger than that of the reflected signal. The amplitude of the coupling control signal may be smaller than that of the reflected signal. The amplitude of the coupling control signal may be the same as that of the reflected signal. The amplitude and phase of the synthetic signal may be adjusted depending on the amplitude of the coupling control signal.

[0080] When a coupling control signal having less than twice the amplitude of the reflected signal is applied, the amplitude of the synthetic signal may become smaller than that of the reflected signal. When the amplitude of the coupling control signal is greater than 0 times and less than 1 times the amplitude of the reflected signal, the synthetic signal may have the same phase as the reflected signal and may have a smaller amplitude than the reflected signal. When the amplitude of the coupling control signal is greater than 1 times and less than 2 times the amplitude of the reflected signal, the synthetic signal may have a phase difference of about 180° relative to the reflected signal and may have a smaller amplitude than the reflected signal. When the amplitude of the coupling control signal exceeds twice the amplitude of the reflected signal, the synthetic signal may have a phase difference of about 180° relative to the reflected signal and may have a larger amplitude than the reflected signal. When the amplitude of the coupling control signal is about the same as that of the reflected signal, the generation of the synthetic signal may not be generated. For example, the coupling control signal and the reflected signal may substantially cancel each other and the synthetic signal may have 0 amplitude. For example, when the waveform of the coupling control signal is about the same as that of the reflected signal and the amplitude of the coupling control signal is about the same as that of the reflected signal, the reflected signal may be removed. In addition, when the amplitude of the coupling control signal is twice the amplitude of the reflected signal, the synthetic signal may have a phase difference of about 180° relative to the reflected signal and may have the same amplitude as the reflected signal.

[0081] FIG. **5** illustrates a relationship between a coupling control signal applied by the edge bias supply portion **40** and a reflected signal according to another embodiment.

[0082] The reflected signal is shown in graph **501**, and the coupling control signal is shown in graph **502**. The horizontal axis represents time, and the vertical axis is voltage, which is the magnitude of the signal.

[0083] Referring to FIG. **5**, the edge bias supply portion **40** may apply a coupling control signal having a phase that constructively interferes with the reflected signal. Constructive interference may occur when different waves interact so that the waves produce a resultant wave with a larger displacement. That is, the edge bias supply portion **40** may apply a coupling control signal having the same phase as the reflected signal (that is, a phase difference of about 0° as indicated by line 0). Accordingly, the coupling control signal may constructively interfere with the reflected signal, and a synthetic signal may be generated by the interaction of the reflected signal and the coupling control signal. The coupling control signal applied to the edge bias electrode **27** may have a larger amplitude than the reflected signal.

[0084] The amplitude of the coupling control signal may be larger than that of the reflected signal. The amplitude of the coupling control signal may be smaller than that of the reflected signal. The amplitude of the coupling control signal may be the same as that of the reflected signal. The amplitude of the synthetic signal may be adjusted depending on the amplitude of the coupling control signal.

[0085] FIG. 6 and FIG. 7 respectively illustrate a relationship between a coupling control signal applied by the edge bias supply portion 40 and a reflected signal according to another embodiment.

[0086] The reflected signal is shown in graph 601, and the coupling control signal is shown in graph 602. The horizontal axis may represent time, and the vertical axis may represent voltage, which may be the magnitude of the signal.

[0087] Referring to FIG. 6, the edge bias supply portion 40 may apply a coupling control signal with a phase difference (B) between more than about 180° and less than about 360° relative to the reflected signal.

[0088] In addition, referring to FIG. 7, the edge bias supply portion 40 may apply a coupling control signal with a phase difference (C) between more than about 0° and less than about 180° relative to the reflected signal.

[0089] The coupling control signal and the reflected signal may be synthesized to generate a synthetic signal. The synthetic signal may have a changed phase compared to the reflected signal. In other words, the synthetic signal may have a phase difference between more than about 0° and less than about 360° relative to the reflected signal.

[0090] The amplitude of the coupling control signal may be larger than that of the reflected signal. The amplitude of the coupling control signal may be smaller than that of the reflected signal. The amplitude of the coupling control signal may be the same as that of the reflected signal. The amplitude and phase of the synthetic signal may be adjusted depending on the amplitude of the coupling control signal.

[0091] FIG. 8 illustrates a relationship between an incident signal and a synthetic signal according to an embodiment.

[0092] Referring to FIG. 8, the synthetic signal generated by some embodiments according to FIG. 4 to FIG. 7 may have a phase that destructively interferes with the incident signal. That is, the synthetic signal generated by some embodiments according to FIG. 4 to FIG. 7 may have a phase difference (D) of about 180° from the incident signal.

[0093] In other words, the synthetic signal may be generated to have a phase difference from the reflected signal. In FIG. 8, the synthetic signal may have a phase difference of about 180° from the incident signal. Accordingly, destructive interference may occur between the incident signal and the synthesized signal.

[0094] In addition, the amplitude of the synthetic signal may be adjusted according to the amplitude of the coupling control signal applied by the edge bias supply portion 40. When the amplitude of the synthetic signal is greater than 0 times and less than 1 times the amplitude of the incident signal, the resultant signal generated from the interaction of the incident signal and the synthetic signal may have the same phase as the incident signal and may have a smaller amplitude than the incident signal. When the amplitude of the synthetic signal is 1 times the amplitude of the incident signal, a resultant signal may not be generated due to complete destructive interference. When the amplitude of the synthetic signal is greater than 1 times and less than 2 times the amplitude of the incident signal, the resultant signal generated from the interaction of the incident signal and the synthetic signal may have a phase difference of about 180° from incident signal and may have a smaller amplitude than the incident signal. When the amplitude of the synthetic signal is twice the amplitude of the incident signal, the resultant signal generated from the interaction of the incident signal and the synthetic signal may have a phase difference of about 180° from incident signal and may have the same amplitude as the incident signal. When the amplitude of the synthetic signal exceeds twice the amplitude of the incident signal, the resultant signal generated

from the interaction of the incident signal and the synthetic signal may have a phase difference of about 180° from incident signal and may have a larger amplitude than the incident signal.

[0095] FIG. **9** illustrates a relationship between an incident signal and a synthetic signal according to another embodiment.

[0096] Referring to FIG. **9**, the synthetic signal generated by some embodiments according to FIG. **4** to FIG. **7** may have a phase that constructively interferes with the incident signal. That is, the synthetic signal generated by some embodiments according to FIG. **4** to FIG. **7** may be in the same phase as the incident signal (that is, a phase difference of 0°).

[0097] In other words, the synthetic signal may be generated to have a phase difference from the reflected signal. In FIG. **9**, the synthetic signal may have the same phase as the incident signal. Accordingly, the incident signal and the synthetic signal may constructively interfere, and a resultant signal that has the same phase as the incident signal and has a larger amplitude than the incident signal may be generated.

[0098] FIG. **10** and FIG. **11** respectively illustrate a relationship between an incident signal and a synthetic signal according to another embodiment.

[0099] Referring to FIG. **10** and FIG. **11**, the synthetic signal generated in some embodiments according to FIG. **4** to FIG. **7** may have a phase difference from the incident signal. That is, as shown in FIG. **10**, the synthetic signal may have a phase difference (E) of more than 180° and less than 360° from the incident signal. In addition, as shown in FIG. **11**, the synthetic signal may have a phase difference (F) of more than 0° and less than 180° from the incident signal.

[0100] Accordingly, the resultant signal applied to the edge bias supply portion **40** by the coupling signal and the coupling control signal may be in a form in which the phase, or the phase and amplitude, may be adjusted compared to the coupling signal.

[0101] FIG. **12** illustrates an edge bias supply portion **40** according to an embodiment.

[0102] Referring to FIG. **12**, the edge bias supply portion **40** may include a coupling control power source **410** and a sheath control power source **420**.

[0103] The coupling control power source **410** may apply the coupling control signal to the edge bias electrode **27**. That is, the coupling control power source **410** may apply the coupling control signal to the edge bias electrode **27** as in some embodiments described herein with reference to FIG. **4** to FIG. **7**.

[0104] The sheath control power source **420** may apply the sheath control signal to the edge bias electrode **27**. The sheath control power source **420** and the coupling control power source **410** may be provided as voltage sources connected to each other in series. The sheath control signal may have a frequency different from that of the coupling control signal. In addition, the sheath control signal may have a waveform different from that of the coupling control signal. For example, the sheath control signal may be a square wave or the like. The frequency of the sheath control power source **420** may be smaller than the frequency of the excitation supply portion **33**.

[0105] According to an embodiment, the edge bias supply portion **40** may allow a signal optimized for controlling the state of the sheath to be applied to the edge bias electrode **27**. That is, the edge bias supply portion **40** may apply a sheath control signal for controlling the sheath separately from the coupling control signal for controlling the coupling signal.

[0106] The coupling signal may be generated even when the edge bias supply portion **40** is in the off state. In a comparative embodiment, the signal applied to the edge bias electrode **27** may be synthesized with the coupling signal, and the signal at the edge bias electrode **27** may become larger than the coupling signal.

[0107] A synthetic signal of the reflected signal and the coupling control signal may be applied to the edge bias electrode **27** by the coupling control signal applied by the edge bias supply portion **40**. In this case, the synthetic signal may have a smaller amplitude or a larger amplitude than the reflected signal. In addition, the synthetic signal may be in a phase-changed state compared to the reflected signal. In addition, the synthetic signal may interfere destructively or constructively with

the incident signal, or may change the phase of the incident signal. Accordingly, when the coupling control signal is applied in a form that substantially cancels the coupling signal, the sheath control signal may be applied to the edge bias electrode **27**. That is, a signal smaller than the coupling signal may ultimately be applied to the edge bias electrode **27**. In addition, a signal having a waveform according to the sheath control signal may be applied to the edge bias electrode **27** with the coupling signal substantially reduced.

[0108] FIG. **13** illustrates an upper portion of a support member **20a** according to another embodiment.

[0109] Referring to FIG. **13**, a support member **20a** may include a support plate **200a**, a focus ring **210a**, a support ring **220a**, and a thermal pad **240**.

[0110] The support plate **200a**, the focus ring **210a**, and the support ring **220a** may be the same as or similar to the support plate **200**, the focus ring **210**, and the support ring **220** described with reference to FIG. **2**, and repeated descriptions thereof may be omitted.

[0111] An edge bias electrode **27a** may be disposed below the focus ring **210a**.

[0112] The edge bias electrode **27a** may be the same as or similar to the edge bias electrode **27** described above in FIG. **1** and FIG. **2**, and repeated descriptions thereof may be omitted.

[0113] An edge bias supply portion **40a**, an impedance control member **42a**, and a measurement member **41a** may be connected to the edge bias electrode **27a**. Since the edge bias supply portion **40a**, the impedance control member **42a**, and the measurement member **41a** may be the same as or similar to the edge bias supply portion **40**, the impedance control member **42**, and the measurement member **41** described with reference to FIG. **1** to FIG. **12**, repeated descriptions thereof may be omitted.

[0114] The thermal pad **240** may be disposed between the focus ring **210a** and the support plate **200a**. The thermal pad **240** may be disposed between a lower surface of the focus ring **210a** and an upper surface of the support plate **200a**. The thermal pad **240** may be made of a material having higher thermal conductivity than the focus ring **210a**. Accordingly, the thermal pad **240** may improve thermal conductivity between the focus ring **210a** and the support plate **200a**. As the thermal conductivity is improved, a temperature difference between the focus ring **210a** and the support plate **200a** may be reduced during the fixing process.

[0115] When the thermal conductivity between the focus ring **210a** and the support plate **200a** increases, the magnitude of the coupling signal generated in the edge bias electrode **27a** may increase. In this case, by adjusting the coupling signal through the coupling control signal supplied by the edge bias supply portion **40a**, a change in the signal applied to the edge bias electrode **27a** by the thermal pad **240** may be reduced.

[0116] FIG. **14** illustrates an upper portion of a support member **20b** according to another embodiment.

[0117] Referring to FIG. **14**, the support member **20b** may include a support plate **200b**, a focus ring **210b**, and a support ring **220b**.

[0118] The support plate **200b**, the focus ring **210b**, and the support ring **220b** may be the same as or similar to the support plate **200**, the focus ring **210**, and the support ring **220** described with reference to FIG. **2**, and repeated descriptions thereof may be omitted.

[0119] In addition, a thermal pad may be disposed between the focus ring **210b** and the support plate **200b** in the same or similar manner as described above in FIG. **13**, and repeated descriptions thereof may be omitted.

[0120] An edge bias electrode **27b** may be disposed inside the focus ring **210b**. In other respects, the edge bias electrode **27b** may be the same as or similar to the edge bias electrode **27** described with reference to FIG. **1** and FIG. **2**, and repeated descriptions thereof will be omitted.

[0121] An edge bias supply portion **40b**, an impedance control member **42b**, and a measurement member **41b** may be connected to the edge bias electrode **27b**. The edge bias supply portion **40b**, the impedance control member **42b**, and the measurement member **41b** may be the same as or

similar to the edge bias supply portion **40**, the impedance control member **42**, and the measurement member **41** described with reference to FIG. **1** to FIG. **12**, and repeated descriptions thereof may be omitted.

[0122] FIG. **15** illustrates an upper portion of a support member **20c** according to another embodiment.

[0123] Referring to FIG. **15**, the support member **20c** may include a support plate **200c**, a focus ring **210c**, and a support ring **220c**.

[0124] The support plate **200c**, the focus ring **210c**, and the support ring **220c** may be the same as or similar to the support plate **200**, the focus ring **210**, and the support ring **220** described with reference to FIG. **2**, and repeated descriptions thereof may be omitted.

[0125] In addition, a thermal pad may be disposed between the focus ring **210c** and the support plate **200c** in the same or similar manner as described above in FIG. **13**, and repeated descriptions thereof may be omitted.

[0126] An edge bias supply portion **40c**, an impedance control member **42c**, and a measurement member **41c** may be connected to the focus ring **210c**. The focus ring **210c** may function as the edge bias electrode **27** described in FIG. **1** to FIG. **12**. The focus ring **210c** may function as an edge bias electrode, the function of the edge bias electrode may be the same as or similar to the edge bias electrode **27** described above in FIG. **1** to FIG. **12**, and repeated descriptions thereof may be omitted.

[0127] The edge bias supply portion **40c**, the impedance control member **42c**, and the measurement member **41c** may be the same as or similar to the edge bias supply portion **40**, the impedance control member **42**, and the measurement member **41** described with reference to FIG. **1** to FIG. **12**, and repeated descriptions thereof may be omitted.

[0128] FIG. **16** illustrates a substrate processing apparatus **1d** according to another embodiment.

[0129] Referring to FIG. **16**, the substrate processing apparatus **1d** according to another embodiment may include a chamber **10d**, a support member **20d**, and a plasma excitation member **50d**.

[0130] The inside of the chamber **10d** may define a process space PSd. A substrate processing process may be performed within the process space PSd.

[0131] The support member **20d** may be disposed inside the chamber **10d**. The support member **20d** may include a fixing electrode **21d**, a heating member **22d**, a lower electrode **23d**, a bias electrode **24d**, and an edge bias electrode **27d**.

[0132] A fixing supply portion **31d**, a heating supply portion **32d**, an excitation supply portion **33d**, a matcher **34d**, a bias supply portion **35d**, an edge bias supply portion **40d**, an impedance control member **42d**, and a measurement member **41d** may be connected to the support member **20d**.

[0133] The lower electrode **23d** may be disposed inside the support member **20d**.

[0134] The bias electrode **24d** may be disposed inside the support member **20d**. The bias supply portion **35d** may be connected to the support member **20d**. The bias supply portion **35d** may be electrically connected to the bias electrode **24d**.

[0135] The chamber **10d** and the plasma excitation member **50d** may be the same as or similar to the substrate processing apparatus **1** of FIG. **1**, and repeated descriptions thereof may be omitted.

[0136] The bias electrode **24d** may be provided separately from the lower electrode **23d**. In other respects, the structure of the support member **20d**, and the fixing supply portion **31d**, the heating supply portion **32d**, the excitation supply portion **33d**, the matcher **34d**, the bias supply portion **35d**, the edge bias supply portion **40d**, the impedance control member **42d**, and the measurement member **41d** that are connected to the support member **20d** may be the same as or similar to some embodiments described with reference to FIG. **1** to FIG. **15**. Accordingly, repeated descriptions thereof may be omitted.

[0137] FIG. **17** illustrates a substrate processing apparatus **1e** according to another embodiment.

[0138] Referring to FIG. **17**, the substrate processing apparatus **1e** according to another

embodiment may include a chamber **10e**, a support member **20e**, and a plasma excitation member **50e**.

[0139] The chamber **10e** may define a process space PSe. A substrate processing process may be performed within the process space PSe. At least a portion of an upper wall of the chamber **10e** may be made of a dielectric material. In other respects, the chamber **10e** may be the same as or similar to those of the chamber **10** of FIG. **1**, and repeated descriptions thereof may be omitted.

[0140] The support member **20e** may be disposed inside the chamber **10e**. The support member **20e** may include a fixing electrode **21e**, a heating member **22e**, a bias electrode **24e**, and an edge bias electrode **27e**.

[0141] A fixing supply portion **31e**, a heating supply portion **32e**, a bias supply portion **45e**, an edge bias supply portion **40e**, an impedance control member **42e**, and a measurement member **41e** may be connected to the support member **20e**.

[0142] The fixing supply portion **31e** may be electrically connected to the fixing electrode **21e**. The heating supply portion **32e** may be electrically connected to the heating member **22e**. The bias supply portion **45e** may be electrically connected to the bias electrode **24e**. The edge bias supply portion **40e**, the impedance control member **42e**, and the measurement member **41e** may be electrically connected to the edge bias electrode **27e**.

[0143] The lower electrode **23** described above in FIG. **1** may be omitted from the support member **20e**. Accordingly, the excitation supply portion **33** and the matcher **34**, which are electrically connected to the lower electrode **23** described with reference to FIG. **1**, may not be connected to the support member **20e**. The lower electrode **23d** described with reference to FIG. **16** may be omitted from the support member **20e**. Accordingly, the excitation supply portion **33d** and the matcher **34d**, which are electrically connected to the lower electrode **23d** described with reference to FIG. **16**, may not be connected to the support member **20e**. The remaining structure of the support member **20e**, and the fixing supply portion **31e**, the heating supply portion **32e**, the bias supply portion **45e**, the edge bias supply portion **40e**, the impedance control member **42e**, and the measurement member **41e** that are connected to the support member **20e** may be the same as or similar to some embodiments described with reference to FIG. **1** to FIG. **16**. Accordingly, repeated descriptions thereof may be omitted.

[0144] The plasma excitation member **50e** may allow energy for excitation of plasma to be applied to the inside of the chamber **10e**. The plasma excitation member **50e** may have an antenna structure. The plasma excitation member **50e** may be disposed outside of the chamber **10e**. The plasma excitation member **50e** may be disposed adjacent to the upper surface of the upper wall of the chamber **10e**. The plasma excitation member **50e** may be disposed to face the inner space of the chamber **10e**, with the upper wall of the chamber **10e** interposed therebetween.

[0145] The plasma excitation member **50e** may be electrically connected to the excitation supply portion **55e**. The excitation supply portion **55e** may be provided as a high-frequency power source that is configured to generate high-frequency power. The excitation supply portion **55e** may be provided as an RF power source. The plasma excitation member **50e** may generate an electromagnetic wave through the power provided by the excitation supply portion **55e**. The gas supplied into the chamber **10e** may be excited by the plasma by the electromagnetic wave generated by the plasma excitation member **50e**.

[0146] While embodiment of the present disclosure has been described in connection with what is presently considered to be practical embodiments, it is to be understood that the disclosure is not limited to example embodiments, but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the spirit and scope of the appended claims.

DESCRIPTION OF SYMBOLS

[0147] **10**: chamber [0148] **20**: support member [0149] **21**: fixing electrode [0150] **22**: heating member [0151] **23**: lower electrode [0152] **27**: edge bias electrode [0153] **31**: fixing supply portion [0154] **32**: heating supply portion [0155] **33**: excitation supply portion [0156] **34**: matcher [0157]

35: bias supply portion [0158] **40:** edge bias supply portion [0159] **41:** measurement member [0160] **42:** impedance control member [0161] **50:** plasma excitation member

Claims

1. A substrate processing apparatus comprising: a chamber; a support member disposed inside the chamber and including a bias electrode and an edge bias electrode; a bias supply portion electrically connected to the bias electrode; and an edge bias supply portion electrically connected to the edge bias electrode, wherein the edge bias supply portion applies a coupling control signal having a frequency corresponding to a coupling signal, and the coupling signal is generated at the edge bias electrode by coupling with a bias signal generated at the bias electrode.
2. The substrate processing apparatus of claim 1, further comprising a measurement member electrically connected to the edge bias electrode, wherein the measurement member measures the coupling signal.
3. The substrate processing apparatus of claim 2, wherein the measurement member is disposed between the edge bias electrode and the edge bias supply portion.
4. The substrate processing apparatus of claim 2, further comprising an impedance control member electrically connected to the edge bias electrode, wherein the impedance control member controls the coupling signal.
5. The substrate processing apparatus of claim 4, wherein the measurement member is disposed between the impedance control member and the edge bias electrode.
6. The substrate processing apparatus of claim 1, wherein the coupling signal includes an incident signal traveling in a direction from the edge bias electrode to the edge bias supply portion, and a reflected signal traveling in a direction from the edge bias supply portion to the edge bias electrode, and wherein the edge bias supply portion applies the coupling control signal having a frequency corresponding to the reflected signal.
7. The substrate processing apparatus of claim 6, wherein the coupling control signal has a phase that destructively interferes with the reflected signal.
8. The substrate processing apparatus of claim 6, wherein the coupling control signal has a phase that constructively interferes with the reflected signal.
9. The substrate processing apparatus of claim 6, wherein a synthetic signal generated by combining the coupling control signal and the reflected signal has a phase difference between more than 0° and less than 360° relative to the reflected signal.
10. The substrate processing apparatus of claim 6, wherein a synthetic signal generated by combining the coupling control signal and the reflected signal has a phase that destructively interferes with the incident signal.
11. The substrate processing apparatus of claim 6, wherein a synthetic signal generated by combining the coupling control signal and the reflected signal has a phase that constructively interferes with the incident signal.
12. The substrate processing apparatus of claim 6, wherein a synthetic signal generated by combining the coupling control signal and the reflected signal has a phase difference between more than 0° and less than 180° relative to the incident signal.
13. The substrate processing apparatus of claim 6, wherein a synthetic signal generated by combining the coupling control signal and the reflected signal has a phase difference between more than 180° and less than 360° relative to the incident signal.
14. A substrate processing apparatus comprising: a chamber; a support member disposed inside the chamber and including a bias electrode and an edge bias electrode; a bias supply portion electrically connected to the bias electrode; an edge bias supply portion electrically connected to the edge bias electrode; and a measurement member electrically connected to the edge bias electrode, wherein the measurement member measures a coupling signal generated at the edge bias

electrode by coupling with a bias signal generated at the bias electrode.

15. The substrate processing apparatus of claim 14, wherein the edge bias supply portion applies a coupling control signal having a frequency corresponding to the coupling signal.

16. The substrate processing apparatus of claim 14, wherein the support member comprises: a support plate; a focus ring disposed on an outer periphery of an upper portion of the support plate; and a support ring disposed below the focus ring, wherein the edge bias electrode is disposed inside the support ring.

17. The substrate processing apparatus of claim 16, wherein the support member comprises: a support plate; a focus ring disposed on an outer periphery of an upper portion of the support plate; and a support ring disposed below the focus ring, wherein the edge bias electrode is disposed inside the focus ring.

18. The substrate processing apparatus of claim 16, wherein the support member comprises: a support plate; a focus ring disposed on an outer periphery of an upper surface of the support plate; and a support ring disposed below the focus ring, wherein the focus ring is electrically connected to the edge bias supply portion and functions as the edge bias electrode.

19. A substrate processing apparatus comprising: a chamber; a support member disposed inside the chamber and including a bias electrode and an edge bias electrode; a bias supply portion electrically connected to the bias electrode; an edge bias supply portion electrically connected to the edge bias electrode; and a measurement member electrically connected to the edge bias electrode, wherein the measurement member measures a coupling signal generated at the edge bias electrode by coupling with a bias signal generated at the bias electrode, and the edge bias supply portion applies a coupling control signal having a frequency corresponding to a reflected signal to generate a synthetic signal in which at least one of an amplitude or a phase of the synthetic signal is different than an amplitude or a phase of the reflected signal, wherein the reflected signal is a portion of the coupling signal and traveling in a direction from the edge bias supply portion toward the edge bias electrode.

20. The substrate processing apparatus of claim 19, wherein the edge bias supply portion comprises: a coupling control power source that applies the coupling control signal; and a sheath control power source that applies a sheath control signal whose frequency or waveform is different from the coupling control signal.
